

## Allison H. Moore – Curriculum Vitae

Department of Mathematics & Applied Mathematics  
Virginia Commonwealth University  
1015 Floyd Avenue | Box 842014  
Richmond, VA 23284-2014 (804) 828-1301

moorea14@vcu.edu  
<https://allisonhmoore.github.io/>

### Education

- Ph.D. in Mathematics, The University of Texas (2013)
- B.S. in Mathematics, with High Honors, The University of Texas (2006)
- B.A. in Plan II with High Honors, The University of Texas (2006)
- Undergraduate Study, Independent University of Moscow, Russia (2004)

### Professional Positions/Appointments

- Associate Professor, Virginia Commonwealth University  
Department of Mathematics & Applied Mathematics July 2024 – current
- Assistant Professor, Virginia Commonwealth University  
Department of Mathematics & Applied Mathematics August 2019 – June 2024
- Krener Assistant Professor, University of California, Davis  
Department of Mathematics July 2016 -- June 2019
- Postdoctoral appointment, University of California, Davis  
Department of Microbiology and Molecular Genetics July 2017-- June 2019
- RTG Lovett Instructor, Rice University,  
Department of Mathematics Fall 2013 -- June 2016
- The University of Texas, Department of Mathematics:
  - Assistant Instructor Fall 2012
  - Research/Teaching Assistant Fall 2006 --- Spring 2013
  - Coordinator of the Saturday Morning Math Group Spring 2011 -- Spring 2012

### Honors and Awards

- 2023, VCU, *National/International Recognition Award (NIRA) Scholar*.
- 2023, VCU, Nominated for the University Early Career Faculty Award. (Nomination only)
- 2018, University of California, Davis. *G. Thomas Sallee Mathematics Teaching Award*, \$500. Excellence in lower division undergraduate mathematics courses.
- 2009, University of Texas. *Frank Gerth III Teaching Excellence Award*, \$1,000.
- 2013, University of Texas. Professional Development Award, \$400.
- 2009 and 2007. University of Texas. Math Department Summer Fellowships.
- 2006. University of Texas Vigre NSF Research Grant for Undergraduates, \$5,000.
- 2004. AMS/NSF Scholarship Grant, *Math in Moscow*, \$5,000.
- 2004. The University of Texas. Plan II Research and Travel Grant, \$1,500.

## RESEARCH

### Publications<sup>1</sup>

#### Peer reviewed publications:

- Tye Lidman and Allison H. Moore. Adjacency of Three-Manifolds and Brunnian Links. *Nagoya Mathematical Journal*. 261:e10. doi:10.1017/nmj.2025.10092, (2026).
- Kenneth Baker, Dorothy Buck, Allison H. Moore, Danielle O'Donnol, and Scott Taylor. Primality of theta-curves with proper rational tangle unknotting number one. *Transactions of the American Mathematical Society, Series B*, 12(08):276–297, (2025).

---

<sup>1</sup> Authorship is alphabetical in Mathematics (1—25). Authorship is non-alphabetical in Education (26—27).

3. Artem Kotelskiy, Tye Lidman, Allison H. Moore, Liam Watson, and Claudius Zibrowius. Cosmetic operations and Khovanov multicurves, *Mathematische Annalen*, 389(3): 2903—2930, (2024).
4. Eugene Gorsky, Beibei Liu, Tye Lidman and Allison H. Moore. Triple Linking Numbers and Heegaard Floer Homology. *International Mathematical Research Notices*, 2023(6): 4501–4554, (2023).
5. Tye Lidman, Allison H. Moore and Claudius Zibrowius. L-space knots have no essential Conway spheres. *Geometry & Topology*, 26(5): 2065–2102, (2022).
6. Stanislav Jabuka, Beibei Liu, and Allison H. Moore. Knot graphs and Gromov hyperbolicity. *Mathematische Zeitschrift*, 301(1):811–834, (2022).
7. Christopher Flippen, Allison H. Moore, and Essak Seddiq. Quotients of the Gordian and  $H(2)$ -Gordian Graphs. *Journal of Knot Theory and Its Ramifications*, 30(05):2150037, (2021).
8. Allison H. Moore and Mariel Vazquez. A note on band surgery and the signature of a knot. *Bulletin of the London Mathematical Society*, 52(6):1191—1208, (2020).
9. Eugene Gorsky, Beibei Liu and Allison H. Moore. Surgery on links of linking number zero and the Heegaard Floer d-invariant. *Quantum Topology* 11(2):323—378, (2020).
10. Allison H. Moore and Mariel Vazquez. The non-coherent band surgery model for site-specific recombination. In *Topology and Geometry of Biopolymers*, volume 746, *Contemporary Mathematics*, pages 101–125. American Mathematical Society, Providence, RI, (2020).
11. Tye Lidman, Allison H. Moore and Mariel Vazquez. Distance one lens space fillings and band surgeries on the trefoil knot. *Algebraic & Geometric Topology*; 19(5):2439--2484, (2019).
12. Kenneth Baker and Allison H. Moore. Montesinos knots, Hopf plumbings, and L-space surgeries. *Journal of the Mathematical Society of Japan*, 70(1):95—110, (2018).
13. Tye Lidman and Allison H. Moore. Cosmetic surgery in L-spaces and nugatory crossings. *Transactions of the American Mathematical Society*, 369(5):3639—3654, (2017).
14. Tye Lidman and Allison H. Moore. Pretzel knots with L-space surgeries. *Michigan Mathematical Journal*, 65(1):105--130, (2016).
15. Allison H. Moore. Symmetric unions without cosmetic crossing changes. In Gail Letzter et al., editors, *Advances in the Mathematical Sciences: Research from the 2015 Association for Women in Mathematics Symposium*, volume 6, pages 103—116. Springer International Publishing, Cham, (2016).
16. Allison H. Moore and Laura Starkston. Genus-two mutant knots with the same dimension in knot Floer and Khovanov homologies. *Algebraic & Geometric Topology*, 15(1):43—63, (2015).

#### Accepted Articles:

17. Matthew Elpers, Rayan Ibrahim, and Allison H. Moore. Determinants of simple theta curves and symmetric graphs. arXiv:2211.00626 [math.GT], Accepted at *Journal of Knot Theory and its Ramifications*. 12 pages.
18. Kenneth L. Baker, Allison H. Moore, Danielle O'Donnol, and Scott Taylor. Signature, slicing foams, and crossing changes of Klein graphs. arXiv:2405.15044 [math.GT], Accepted at *Algebraic & Geometric Topology*. 23 pages.
19. Allison H. Moore and Nicola Tarasca. On gluing and splitting series invariants of plumbed 3-manifolds. arXiv:2506.06515 [math.GT], 29 pages, Accepted by the *Journal of Geometry and Physics*.
20. Rayan Ibrahim and Allison H. Moore. Methods for Analyzing RNA Pseudoknots via Chord Diagrams and Intersection Graphs. arXiv:2512.19939 [q-bio.BM], 26 pages. Accepted by the *Bulletin of Mathematical Biology*.

#### Submitted and Preprints:

21. Allison H. Moore and Nicola Tarasca. Root lattices and invariant series for plumbed 3-manifolds. arXiv:2405.14972 [math.GT], (2024). 38 pages. Submitted.

#### Thesis:

22. Allison H. Moore. Behavior of Knot Floer Homology Under Conway and Genus Two Mutation. *PhD Thesis*, Department of Mathematics, University of Texas at Austin, (2013).

### Software and Websites:

23. Allison H. Moore, Milad Nasrollahi and Shawn Witte. *BraidGenerator* (Software), (2019). URL: <https://pypi.org/project/braidgenerator/>
24. C. Livingston and A. H. Moore, *KnotInfo: Table of Knot Invariants*, URL: [knotinfo.math.indiana.edu](http://knotinfo.math.indiana.edu)
25. C. Livingston and A. H. Moore, *LinkInfo: Table of Link Invariants*, URL: [linkinfo.sitehost.iu.edu](http://linkinfo.sitehost.iu.edu)

### Scholarship of Teaching and Learning Articles:

26. [Peer reviewed conference proceeding] Bontrager, B. J., Spring, A., Satyam, V. R., Bae, C. L., Aldi, M., Moore, A. H., Tarasca, N., & Freeman, H. (2024). Instructional moves to support middle school students' engagement during a geometry camp. In K. W. Kosko, J. Caniglia, S. A. Courtney, M. Zolfaghari, & G. A. Morris (Eds.), *Proceedings of the forty-sixth annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (pp. 1767-1772). Kent State University.
27. [Book chapter] Bontrager, B., Spring, A., Satyam, V. R., Bae, C. L., Aldi, M., Tarasca, N., & Moore, A. (in press). Middle school students' emotional experiences in mathematics represented through emotion graphs: A multimodal discourse analysis. In L. de Oliveira (Ed.), *Discourse Analysis in Practice* (pp. 11-30). Emerald Publishing Limited.

### Grants and Funding Awarded

- NSF DMS— 2204148. *Explorations in Entanglement and Knotting in Low-Dimensional Topology*. 8/01/2022—7/31/2025, \$294,584. (Sole PI)
- NSF DMS— 2528113. *Conference: Richmond Geometry Festival: New Trends in Homological Invariants*. 8/1/2025 – 7/31/2026. \$29,276. (Co-PIs Aldi, Tarasca)
- NSF DMS— 2349810. *Conference: Richmond Geometry Festival: Geometric Topology and Moduli*. 5/1/2024 – 10/31/2025. \$26,430. (Co-PI with Aldi, Tarasca)
- NSF DMS— 2240741. *Conference: Richmond Geometry Festival: Knots, Moduli, and Strings*. 5/15/2023 – 8/15/2023. \$31,108. (Co-PI with Aldi, Tarasca)
- 4-VA Collaborative Research Grant Award (University of Virginia). \$35,000 Co-PIs with Bray, Field, Lukyanenko, Maloni, Quigley. "Collaborative workshops in topology" to be held in 2025-2026.
- VCU Breakthroughs Fund Award. *Symmetry, Surfaces, and Knots: Empowering Middle School Students*. 7/30/2022—7/30/2024, \$200,000. (Co-PI with Aldi, Tarasca)
- VCU Quest Fund Award. *Quantum Fields and Knots: An Integrative Approach*. 7/01/2022—12/31/2023, \$47,794. (Co-PI)
- Jeffress Trust Awards Program in Interdisciplinary Research—FP00014984. *Advancing theory and exploring applications of unknotting to molecular biology* August 2021 – April 2023, \$100,000. (Sole PI).
- VCU HSURP Research Project. *Knots and Links in Three-Manifolds*. Summer 2020, \$600. (Sole PI).

### Other External Support and Funded Activities

- *Summer Research in Mathematics*, Simons Laufer Mathematical Sciences Institute, June 2024. Fully funded resident participant and member of research team in math biology; recipient of childcare grant.
- American Institute of Mathematics SQuaRE, *Unknotting spatial graphs*. Travel funding for five-member collaborative research project in spatial graphs. 2019-2023.
- NSF DMS—1716987. *The Dynamic Genome: Studying the Interplay between Local Strand-Passage and Reconnection*. 7/18/2017—6/30/2019, \$290,000.  
Role: I co-designed the aims of the grant and co-wrote the proposal while at UC Davis with the PI (M. Vazquez). Grant provided 50% of my postdoctoral funding for two years, travel and equipment.
- NSF DMS—1148609. *RTG: Analysis, Geometry, and Topology at Rice University*, Fall 2013--Spring 2016. Role: Postdoctoral salary/travel only; no role in proposal preparation.
- NSF DMS—1148609. *RTG: Unified Training in Geometry and Topology*, Fall 2010, Summer 2011. Role: Grad student support only; no role in proposal preparation.

## **Presentations** (From 2013 – current, for brevity)

### **Minicourses:**

1. *Applied Topology: DNA Topology*. (Mini Course) Canadian Mathematical Society Summer Meeting, Saskatoon, Canada, May 31, 2024.

### **Invited Conference Talks:**

2. *Root lattices and series valued invariants of plumbed 3-manifolds*. Geometric Topology Session, 59th Spring Topology and Dynamics Conference. University of Alabama, Birmingham, March 2026.
3. (i) *Series valued invariants and plumbed 3-manifolds* and (ii) *Discussion/Open Problem Session on KnotInfo and LinkInfo*. Links in Dimensions 3 and 4. ICERM, Providence, RI, May 2025
4. *Series valued invariants and plumbed 3-manifolds*. Gauge Theory and Floer Homology in Low-Dimensional Topology. Simons Center for Geometry and Physics, Stony Brook, NY Apr. 2025
5. *Root lattice puzzles and plumbed 3-manifolds*. Interactions of Geometry and Quantum Topology focused on Links in Thickened Surfaces. Banff International Research Station, Banff, Canada (Online) Apr. 2025.
6. *Tricks with Tangles*. What's your trick? A Non-Traditional Conference in Low-Dimensional Topology, Banff International Research Station, Banff, Canada, Aug. 2024.
7. *Unknotting numbers and invariants of trivalent spatial graphs*. Applied Topology: DNA topology, Material Science, Topological Data Analysis. Canadian Mathematical Society Summer Meeting, Saskatoon, Canada, June 2024.
8. *Unknotting operations and invariants of trivalent spatial graphs*. The Crossroads of Topology, Combinatorics and Biosciences: Deciphering the Entanglement of Multi-Stranded Nucleic Acids. BIRS, Banff, Canada, March 2024.
9. *Unknotting and Invariants of Trivalent Spatial Graphs*. AMS Special Session on Mathematical Modeling of Nucleic Acid Structures. Joint Mathematics Meetings, San Francisco, CA, Jan. 2024
10. *Unknotting in Spatial Trivalent Graphs*. AMS Special Session on Recent Advances in Low-dimensional and Quantum Topology, Mobile, AL, October 2023
11. *A Tale of Two Tangles*. Tangled in Knot Theory, ICERM, Brown University, May 2023
12. *Tangle Decompositions and Immersed Curves*, Low Dimensional Topology and Circle-valued Morse Functions, IMSA, University of Miami, Feb. 2023.
13. *Invariants and unknotting in theta curves*, 8th Annual Applied Geometry and Topology Workshops in Mexico (Online), November 2022.
14. *Invariants and unknotting in theta curves*, AMS Special Session on Topology and Geometry of Multi-Stranded Nucleic Acids, Salt Lake City, UT, October 2022
15. Flash Talk: *Unknotting in spatial theta curves*, 4-variétés: à travers les dimensions, CIRM, Marseille, France, Sep. 2022
16. *Cosmetic crossings, cosmetic surgery and Conway spheres*. Canadian Mathematical Society Winter Meeting, Dec. 2021
17. *Cosmetic crossings, cosmetic surgery and Conway spheres*. Knots in Washington 49.5, Dec. 2021
18. *Tangle decompositions and immersed curves*. Boston Graduate Topology Seminar (regional conference), MIT, Boston MA, Nov. 2021
19. *Knot Graphs and Gromov Hyperbolicity*, Lightning Talk (Online), 4D Topology ARC, American Institute of Mathematics, Mar. 2021
20. *Essential Conway spheres and Floer homology via immersed curves*, Joint Meetings of the AMS/MAA, Special Session on Geometry and Topology in Dimensions 3&4 (Online), Jan. 2021
21. *Essential Conway spheres and Floer homology via immersed curves*, AMS Fall Southeastern Sectional Meeting, Special Session on Applied Knot Theory (Online), Oct. 2020
22. *Surgery, higher-order linking and Heegaard Floer homology*. Mini Symposium: Knot Theory on Okinawa, Okinawa Institute of Science and Technology, Okinawa, Japan, Feb. 2020
23. *Knot graphs and Gromov hyperbolicity*. AMS Special Session on Applications and Computations in Knot Theory, Joint Meetings of the AMS/MAA, Denver, CO, Jan. 2020

24. *Heegaard Floer invariants, surgery and Milnor Numbers*. AMS Sectional on Floer homology in Dimensions 3 and 4, University of Wisconsin-Madison, Madison, WI, Sep. 2019
25. *Site-specific recombination and the band surgery along knots*. The Topology of Nucleic Acids: Research at the Interface of Low-Dimensional Topology, Polymer Physics and Molecular Biology, Banff International Research Center, Banff, Canada, Mar. 2019
26. *Surgery on links and the d-invariant*. Winter Meeting of the Canadian Mathematical Society, Session on Topology, Vancouver, BC, Canada, Dec. 2018
27. *The band surgery model for site specific recombination along circular DNA molecules*. AMS Sectional on Mathematical Methods for the Study of the 3-Dimensional Structure of Biopolymers, SFSU, San Francisco, CA, Nov. 2018
28. *Distance one lens space fillings and band surgery*. Topology in Dimensions 3, 3.5 and 4, University of California, Berkeley, CA, May 2018
29. *Knots, bands and reconnection in circular DNA*. Eastern IL Integrated Conference in Geometry, Dynamics, and Topology, Charleston, IL, Apr. 2018
30. *Arsuaga-Vazquez Workshop Tutorials*. Summer School on Modern Knot Theory: Aspects in Algebra, Analysis, Biology, and Physics. University of Freiburg, Freiburg, Germany, Jun. 2017
31. *Cosmetic surgery in L-spaces and nugatory crossings*. Tech Topology Conference, Georgia Tech, Atlanta, GA, Dec. 2015
32. *Nugatory crossings and symmetric unions*. AMS Special Session on Knots, Links and 3-Manifolds, Rutgers, NJ, Nov. 2015
33. *Cosmetic crossing changes in thin knots*. AWM Research Symposium, University of Maryland, Baltimore, MD, Apr. 2015
34. *Cosmetic crossing changes in knots*. AMS Sectional on Knot Theory and Floer-Type Invariants, East Lansing, MI, Mar. 2015
35. *Montesinos knots, Hopf plumbings and L-space surgeries*. AMS Sectional on Interactions between Knots and Manifolds, San Francisco, CA, Oct. 2014
36. *Montesinos knots and contact structures*. AMS Sectional on Invariants in Low-Dimensional Topology, Baltimore, MD, Mar. 2014
37. *Pretzel knots admitting L-space surgeries*. AMS Sectional on Geometric Topology of Knots and 3-Manifolds, Philadelphia, PA, Oct. 2013

#### **Invited Colloquia:**

38. *Knotted graphs and graphs of knots: how combinatorics and knot theory inform each other*. Runnals Colloquium. Colby College, Waterville ME, Feb. 2026
39. *Cutting Knots Down to Size: The Utility of Tangle Decompositions in Topology*. RIMS (Research Innovation in the Mathematical Sciences). University of Arizona, Tempe, AZ. Jan. 2026.
40. *Knotted graphs and graphs of knots: How combinatorics and knot theory inform each other*. In the REU Colloquium, University of Virginia, Charlottesville, VA, July. 2024
41. *Knotted graphs and graphs of knots: How combinatorics and knot theory inform each other*. University of Richmond, Richmond, VA, Oct. 2023
42. *Knots, bands and reconnection in circular DNA*. Colloquium, Virginia Commonwealth University, Richmond, VA, Dec. 2018
43. *Knots, bands and reconnection in circular DNA*. Colloquium, Western Washington University, Bellingham, WA, Nov. 2018
44. *Knots, bands and reconnection in circular DNA*. Colloquium, University of Nevada at Reno, Reno, NV, Oct. 2018
45. *Band surgery along the trefoil knot with applications to site-specific recombination on circular DNA*. Colloquium, Texas State University, San Marcos, TX, Jan. 2018
46. *Site-specific recombination in circular DNA and band surgery along the trefoil knot*. Colloquium, Towson University, Towson, MD, Dec. 2017
47. *Nugatory crossings and cosmetic surgery in L-spaces*. Colloquium, University of Alabama, Tuscaloosa, AL, Feb. 2016

48. *Montesinos knots and L-space surgeries*. Colloquium, Sam Houston State University, Huntsville, TX, Oct. 2014
49. *Mutation and total dimension in knot Heegaard Floer homology*. Colloquium, Rice University, Houston, TX, Sep. 2013
50. *Mutation and knot Heegaard Floer homology*. Colloquium, University of Nevada at Reno, Reno, NV, May 2013

#### **Invited Seminars:**

51. *The Utility of Tangle Decompositions*. Tetrahedral Geometry/Topology Seminar. Elizabethtown College, Elizabethtown PA, Mar. 2026
52. *Plumbed 3-manifolds and puzzles on root lattices*, Geometry Seminar, University of Virginia, Charlottesville, VA, Apr. 2025
53. *Entanglement and invariants of theta-curves*, Seminar GEOTOP-A (Online), Nov. 2023.
54. *Immersed curves in Floer homology and Conway spheres*, Topology and Geometry Seminar, MIT, Boston, MA, Nov. 2022.
55. *Cosmetic crossings*, Geometry and Topology Seminar, Le Centre de recherche en géométrie et topologie (CIRGET), Université du Québec à Montréal (Online), Apr. 2022.
56. *Conway spheres and immersed curves*. Geometry Seminar, University of Virginia, Charlottesville VA, Nov. 2021
57. *Essential Conway spheres and Floer homology via immersed curves*. Topology Seminar, Boston College, Boston MA, Nov. 2021
58. *Conway spheres and immersed curves*. Topology Seminar, Oklahoma State (Online). Nov. 2021
59. *Essential Conway spheres and Floer homology via immersed curves*, University of Texas Topology Seminar (Online) Nov. 2020
60. *Triple linking and Heegaard Floer homology*, Stanford Topology Seminar (Online) Sep. 2020
61. *Triple linking and Heegaard Floer homology*, Brandeis Topology Seminar (Online) Sep. 2020
62. *Triple linking and Heegaard Floer homology*, GA Tech Topology Seminar (Online), Aug. 2020
63. *Triple linking and Heegaard Floer homology*, The Ohio State University, Seminar on Classical Knots + Virtual Knots (Online) Aug. 2020
64. *Surgery, higher-order linking and Heegaard Floer homology*. University of British Columbia, PIMS/CRG Seminar (Online) May. 2020
65. *Heegaard Floer invariants, integral surgery and higher order linking*. Topology Seminar, Boston College, Boston, MA, Oct. 2019
66. *Heegaard Floer d-invariants and integral surgery*. Topology Seminar, Indiana University, IN, Jun. 2019
67. *Heegaard Floer d-invariants and integral surgery*. Topology Seminar, University of Oregon, Eugene, OR, May 2019
68. *Heegaard Floer d-invariants and integral surgery*. Geometry Seminar, University of Virginia, Charlottesville, VA, Apr. 2019
69. *Distance one lens space fillings and band surgery*. LA Joint Topology Seminar, UCLA, Los Angeles, CA, Apr. 2018
70. (1) *Cosmetic crossing changes* (2) *Symmetric unions without cosmetic crossing changes*, Topology Seminar(s), University of California, Berkeley, CA, Apr. 2017
71. (1) *Cosmetic crossing changes* (2) *Symmetric unions without cosmetic crossing changes*, Geometry and Topology Seminar(s), UC Davis, Davis, CA, Sep. 2016
72. *Cosmetic surgery in L-spaces and nugatory crossings*. Geometry and Topology Seminar, MSU, East Lansing, MI, Nov. 2015
73. *Cosmetic surgery in L-spaces and nugatory crossings*. Topology Seminar, UT Austin, TX, Sep. 2015
74. *Cosmetic surgery in L-spaces and nugatory crossings*. Topology Seminar, Rice, Houston, TX, Sep. 2015
75. *Cosmetic crossing changes in thin knots*. Seminar in Symplectic Geometry, Gauge Theory and Categorification Columbia University, NY, May 2015

76. *Pretzel knots admitting L-space surgeries and questions about mutation*. Geometry and Topology Seminar, Caltech, Pasadena, CA, Nov. 2013
77. *Pretzel knots admitting L-space surgeries*. Geometry and Physics Seminar, University of Miami, Miami, FL, Nov. 2013
78. *Pretzels knots admitting L-space surgeries*. Topology Seminar, Rice University, Houston, TX, Sep. 2013

### **Contributed and Selected Conferences**

79. (Contributed) *Root puzzles and plumbed 3-manifolds*. Joint Mathematics Meetings, Seattle, WA, Jan. 2025
80. (Selected) *Essential Conway spheres and Floer homology via immersed curves*, Workshop in Geometry Topology (Online), Texas Christian University, Jun. 2021
81. (Contributed) *The cosmetic crossing conjecture and nugatory crossings*. Joint Meetings, Seattle, WA, Jan. 2016
82. (Selected) *Cosmetic surgery in L-spaces and nugatory crossings*. Workshop in Geometric Topology, TCU, Fort Worth, TX, Jun. 2015
83. (Selected) *Cosmetic crossing changes in thin knots*. Moab Topology Conference, UT, May 2015
84. (Selected) *Pretzel knots admitting L-space surgeries*. Rolfsenfest, CIRM, Luminy, France, Jul. 2013.
85. (Contributed) *The total rank conjecture in knot Heegaard Floer homology*. Joint Meetings, San Diego, CA, Jan. 2013

### **Other Seminars:**

86. *Tait graphs & determinants of knots and spatial graphs*. Discrete MathemaTea, VCU, Oct. 2023.
87. *Root puzzles and plumbed 3-manifolds*. VCU Discrete Mathematics Seminar, VCU, Mar. 2025
88. *Introduction to Knot Floer Homology Parts 1 & 2*. VCU, Apr. 2023.
89. *Intro to Khovanov Homology (parts 1--3)* Geometry & Topology Seminar (Online), VCU, Apr. 2020
90. *Intro to Jones polynomial: skein theory, Kauffman bracket versus Jones' perspective via von Neumann algebras*. Geometry & Topology Seminar, VCU, Sep. 2019
91. *Temperley Lieb algebras*, Geometry & Topology Seminar, VCU, Sep. 2019
92. *Gordian graphs of knots*. ALPS Seminar, VCU, Sep. 2019
93. *Behavior of knot Floer homology under Conway and genus two mutation*. Topology Seminar (Thesis Defense), University of Texas, Austin, TX, Apr. 2013

## **TEACHING**

### **Courses Taught as Instructor of Record**

|                                                         |                         |
|---------------------------------------------------------|-------------------------|
| Math 697, Directed Research, VCU                        | F21 -- F23              |
| Math 610, Graduate Advanced Linear Algebra, VCU         | S26                     |
| Math 602, Graduate Abstract Algebra II, VCU             | S22                     |
| Math 502, Graduate Abstract Algebra I, VCU              | F21                     |
| Math 490, Mathematical Expositions, VCU                 | S23, F23                |
| Math 411, Excursions in Geometry, VCU                   | F23                     |
| Math 409, General Topology, VCU                         | F22                     |
| Math 401, Abstract Algebra, VCU                         | F20, F24                |
| Math 310, Linear Algebra, VCU                           | F19, F20, S21, S22, S25 |
| Math 307, Vector Calculus, VCU                          | S21, S24                |
| Math 200, Calculus Analytic Geometry 1, VCU             | S20, F24                |
| Math 170, The Language of Artificial Intelligence, VCU  | F25                     |
| Mat 22A, Linear Algebra, UC Davis                       | S17, F17, F18           |
| Mat 141, Euclidean and Non-Euclidean Geometry, UC Davis | W17                     |
| Mat 116A, Short Calculus, UC Davis                      | F16                     |
| Math 306, Elements of Abstract Algebra, Rice University | S16                     |
| Math 111, Differential Calculus, Rice University        | F15                     |

|                                                            |              |
|------------------------------------------------------------|--------------|
| Math 445/540, Graduate Algebraic Topology, Rice University | S15          |
| Math 499, Undergraduate Research Seminar, Rice             | F14—S16      |
| Math 591, Graduate Teaching Seminar, Rice                  | AY 2014—2016 |
| Math 699, Topology Seminar, Rice                           | AY 2014—2016 |
| Math 112, Integral Calculus, Rice University               | S14          |
| Math 368, Graph Theory and Combinatorics, Rice             | F13, F14     |
| M408K, Differential Calculus, University of Texas          | F12          |

### **Courses taught as a Teaching Assistant at the University of Texas:**

*Recitations (6 hours weekly) with office hours (3 hours) and grading:*

|                                                     |                    |
|-----------------------------------------------------|--------------------|
| M408M, Vector Calculus                              | F2008, S2009       |
| M408L, Integral Calculus                            | S2008, Summer 2008 |
| M408K, Differential Calculus                        | F2007              |
| M408C, Sequence, Series, and Multivariable Calculus | F2006, S2007       |

*Office hours and grading only:*

|                                    |       |
|------------------------------------|-------|
| M382C, Graduate Algebraic Topology | F2009 |
| M346, Applied Linear Algebra       | F2009 |

### **New Classes Introduced:**

- Math 411: Excursions in Geometry: new required course for math majors.
- Math 170: The Language of AI: general education math; CHS AI Minor & BIS Practical AI+ program.

### **Instructional Materials**

- Math 411 (Excursions in Geometry): New Course Complete lecture notes made available to students.
- Math 502/602 (Graduate Abstract Algebra I and II): Complete lecture notes made available to students.
- Math 610 (Graduate Advanced Linear Algebra): Complete lecture notes made available to students.
- Math 310 (Linear Algebra): Videos (56 tutorials); learning modules and complete lectures (25 PDFs)
- Math 401 (Abstract Algebra): Complete lecture notes made available to students (34 PDFs)
- Math 170 (Language of AI) Lecture notes, and complete set of worksheets and activities.

### **Graduate Students**

- Rayan Ibrahim. Ph.D. in Systems, Modeling and Analysis, VCU. Graduated August 2024; currently a Visiting Assistant Professor at Lafayette College.
- John Carney. Masters in Mathematics, VCU. Graduated August 2024
- Jonathan Gerhard. Masters in Mathematics, VCU. Degree expected May 2027.

### **Sponsorship of Undergraduate Research**

- 2024-2026. Supervised 4 undergrads RAs on NSF-funded projects in knot theory, combinatorics and mathematical biology.
- 2022-2024. Supervised student (as both undergrad and Master's student) on a PerQ, Jeffress and NSF-funded projects in knot theory resulting in article submitted to math journal.
- 2021-2023. Supervised 5 undergrads RAs on various Jeffress Trust-funded projects in knot theory and graph theory. One project resulted in math journal publication.
- Summer 2020. Honors Summer Undergraduate Research Program (HSURP). Supervised two undergrads on a project on "Knots and Links in Three-Manifolds" resulting in math journal publication.
- Spring 2019. Advised 1 undergrad and 1 grad in computational knot theory project resulting in the Markov-chain based software *BraidGenerator*. Advised 1 undergrad on project in knot theory.
- Summer 2018. *Knot invariants and neural knots*, UC Davis. Supervised 1 undergrad in the construction of a feed-forward neural network to predict knot invariants.
- Spring 2016. *TDA & Low-Dim Topology*, Rice University. 7 students.
- Fall 2015. *Combinatorial knot theory*, Rice University. 12 students.



- Spring 2015. *Combinatorial knot theory*, Rice University. 7 students.
- Fall 2014. *Topological data analysis*, Rice University. 7 students.

### **Other Teaching Activities and Initiatives**

- VCU Twists, Turns and Trajectories Camp, June 2-6, 2025. Math, stats and physics camp for middle school students. Teacher and co-organizer.
- VCU Geometry Camp, June 12-16, 2023. Math camp and IRB-approved education study for middle school students. Co-PI, teacher, and co-organizer.
- VCU Knots and Graphs, May 2022 – Co-organizer of an informal REU at VCU.
- Seminar Co-organizer, *Geometry Student Seminar*, VCU, Fall 2022.
- Seminar Co-organizer, *Teaching Seminar*, Rice University, Fall 2014 -- Spring 2016.
- Teacher, *RTG Writing Seminar*, Rice University, Fall 2014.
- Mentor, RTG Directed Reading Program, University of Texas, Spring 2013.
- Co-organizer, TA Training Workshop, University of Texas, Summer 2012.
- Supplemental Instruction Training, University of Texas, Fall 2007-- Spring 2009.

### **SERVICE**

#### **Service to the Profession**

##### **Referee Work & Expert Opinions:**

- |                                                      |                                                                                 |
|------------------------------------------------------|---------------------------------------------------------------------------------|
| • <i>Algebraic &amp; Geometric Topology</i>          | • <i>Mathematical Research Letters</i>                                          |
| • <i>Advances in Mathematics</i>                     | • <i>Michigan Math Journal</i>                                                  |
| • <i>American Mathematical Monthly</i>               | • <i>Pacific Journal of Mathematics</i>                                         |
| • <i>BIRS Workshop Proposals</i>                     | • <i>Proc. of the American Math. Society</i>                                    |
| • <i>Bulletin of the London Mathematical Society</i> | • <i>Proc. of the Edinburgh Math. Society</i>                                   |
| • <i>Cambridge Philosophical Society</i>             | • <i>Proc. of the London Math. Society</i>                                      |
| • <i>Canadian Journal of Math</i>                    | • <i>Quantum Topology</i>                                                       |
| • <i>Compositio Mathematica</i>                      | • <i>Selecta Mathematica</i>                                                    |
| • <i>Geometriae Dedicata</i>                         | • <i>SIGMA (Symmetry, Integrability and Geometry: Methods and Applications)</i> |
| • <i>Geometry &amp; Topology Monographs</i>          | • <i>Springer Conference Proceedings</i>                                        |
| • <i>Involve: A Journal of Mathematics</i>           | • <i>Transactions of the American Mathematical Society</i>                      |
| • <i>Math. Proc. Cambridge Philosophical Society</i> |                                                                                 |

**NSF Review Panels:** Served on three panels in 2024, 2023, 2020.

##### **Conference and Workshop Co-Organization:**

- *Knots, Groups, and Manifolds*. Centre de Recherches Mathématiques, UQAM, Montreal, August 11-15, 2025.
- *4-VA Collaborative Research Grant Workshops*. To be held in Virginia in summer 2025—2026.
- *Applied Topology: DNA topology, Material Science, Topological Data Analysis Session and Mini Course*. Canadian Mathematical Society Summer Meeting, Saskatoon, Canada, May 2024.
- *Richmond Geometry Day*, VCU (*To be held*) February 7, 2026.
- *Richmond Geometry Festival V*, VCU. September 19-21, 2025.
- *Richmond Geometry Festival IV*, VCU. August 12-14, 2024.
- *Richmond Geometry Festival III*, VCU. June 2-4, 2023.
- *Richmond Geometry Festival II*, VCU. (Virtual). May 27-28, 2022.
- *Richmond Geometry Festival I*, VCU (Virtual). June 10-11, 2021.
- *AMS Special Session on Knots and Links in Low-Dimensional Topology*, AMS Sectional Meeting, University of Virginia, \*cancelled due to Covid.
- *AMS Special Session Developments in Spatial Graphs I and II*, Joint Meetings of AMS/MAA Online, January 6, 2021

- Co-leader (with J. Hom) of 8-member team. *Women in Symplectic and Contact Geometry and Topology Workshop* (WiSCon), ICERM, Providence, RI, July 2019.
- Conference Tutor. *Summer School on Modern Knot Theory: Aspects in Algebra, Analysis, Biology, and Physics*. University of Freiburg, Freiburg, Germany, June 2017.
- *Topology in Dimension 3.5: A conference in memory of Tim Cochran*, Rice University, June 2016
- *AMS Special Session on Geometric Perspectives in Knot Theory*, AMS Sectional Meeting, Loyola University, Chicago, October 2015.

#### Other Service to the Profession:

- Develops and maintains *KnotInfo* with C. Livingston. KnotInfo: Table of Knot Invariants. Free online research database. <http://knotinfo.org/>
- Co-organizer of *Virtual Low-Dimensional Topology*. Online community that hosted a seminar, virtual office hours, and events for the topology community during Covid-19. <https://sites.google.com/bc.edu/virtualtopology/home>
- Panelist in career panel at 3<sup>rd</sup> Richmond Geometry Meeting at VCU, June 2023.
- Panelist in career panel at 4<sup>th</sup> Annual SCMB Symposium in Math-Bio, Dec 2021.
- Panelist in career panel at Eastern IL Integrated Conference in Geometry, Dynamics, and Topology, Charleston, IL, Apr. 2018
- Panelist in grad student panel at NCUWM in 2012.

Professional memberships: American Mathematical Society, Association for Women in Math (past)

### Service to the University (2013 – current)

#### University and College Level Committees and Activities

- AAC&U Institute on AI, Pedagogy, and Curriculum team, F24—S25
- VCU CHS *Faculty Learning Community on Mentoring of Students in Research*, S24
- VCU CHS Grants Reviewer, S23.
- VCU CHS Awards, Leave & Grants Review; Ad Hoc Committee, F23—S24; F25
- VCU OVPRI Proposal Reviewer, S24

#### Seminar Organization

- Geometry and Topology Seminar, VCU, F19—current
- Colloquium, VCU, F21—S22, F25
- ALPS Seminar, VCU, F20—S21, F22—S23
- Topology & Working Topology Seminars, Rice University, F14—S16
- Heegaard Floer 'Computational,' Rice University, F14—S15
- Teaching Seminar, Rice University, F14 – S16.
- RTG Writing Seminar, Rice University, Fall 2014.

#### Math Department Committees and Activities

- *Undergraduate Research*, VCU F24—F25
- *Open House*, VCU F24—F25
- *Math Circle & Outreach*, VCU F24—F25
- *Graduate Affairs*, VCU, F23—S25
- *Undergraduate Curriculum*, VCU, F20—S22
- *Undergraduate Affairs*, VCU F25
- *Analysis Coordinator*, VCU, F22—S23, F24—S25

- *Undergraduate Research*, VCU, F22—F23
- *Website*, VCU, F21—S22
- *Covid Prep Team*, VCU, (Voluntary summer committee), Summer 2020.
- *Colloquium Committee*, Rice, F13--S14.
- *Distinguished Women in Mathematics*, University of Texas, F10--S13.

#### Letters of Recommendation and Career Mentorship:

- Postdoctoral Job & Fellowship Letters: 3 individuals (2023-2025)
- VCU Students: **Numerous** letters for many individuals for the following types of programs:
  - Postdoctoral and faculty positions
  - Ph.D. graduate programs
  - M.S. graduate programs
  - HSURP program
  - Scholarships and fellowships
  - Financial aid appeal
  - REUs & summer/internships
  - Transfer programs
  - Job application references
- UC Davis Students: Numerous letters for graduate programs, transfer programs, scholarship recommendations, and other. (2016—2019)
- Rice University Students: Several letters for graduate Ph.D. programs/fellowships, lecturer position (2015—2016)
- USCIS: National interest waiver petitions for 3 individuals (2019, 2021, 2023).
- Other: One letter of support for a postdoctoral scholar award (2020). One letter of support for sabbatical visitation (2021).
- Career Mentor: 3 VCU Grad Students; Numerous prior at UC Davis, Rice.
- Mentor, RTG Directed Reading Program, University of Texas, Spring 2013.

#### Service to the Community

- Outreach: Career Day at Bon Air Elementary, May 2023.
- VCU Math Circle Activity Leader: April 2022 and Dec. 2022.
- Math Camp Colloquium Speaker, Texas State University, July 2018.
- M-PACT middle school outreach volunteer, UC Davis, April 2018.
- STEM for Girls Workshop volunteer, UC Davis, April 2017
- Math Department Picnic Day volunteer, UC Davis, April 2017
- Faculty sponsor of the Association for Women in Math, Rice U Chapter, 2013 – 2016.
- REU Activity Leader, Sam Houston State University, July 2015
- Girls Exploring Math and Science, Houston Museum of Natural Science, February 2014.
- Panelist & Co-organizer; Sonya Kovalevsky Day, University of Texas, S13, S14.
- Coordinator/Speaker, Saturday Morning Math Group/Sunday Math Circle, 2010—2012.
- Sponsor, AMC and AIME Exams, University of Texas, F11- F12